



Department of Computer Science And Engineering (CSE)

Faculty of Science and Information Technology

Semester : Spring/Summer/Fall Year : 2026

ASSIGNMENT

Course Title : Operating Systems
Course Code : CSE323
Assignment On : Syllabus of Final

Submitted By	Submitted To
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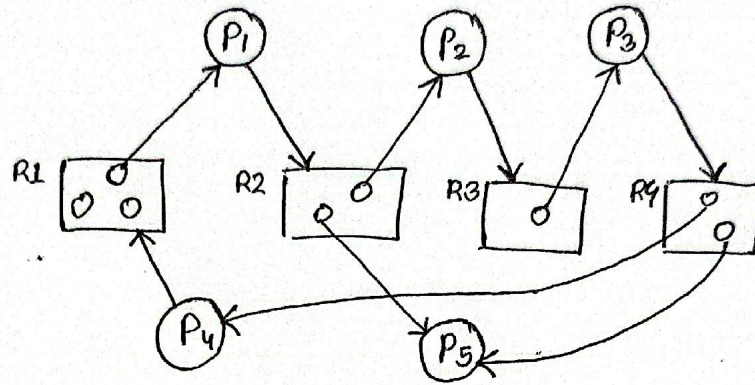
Comment by Course Teacher:

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Operating Systems Assignment

Ans: to the question no-01

Given that,



① Converted it into matrix representation :

Process	Allocation Matrix				Request Matrix				Available Resource (Total - Allocated)			
	R1	R2	R3	R4	R1	R2	R3	R4	R1	R2	R3	R4
P1	1	0	0	0	0	1	0	0	2	0	0	0
P2	0	1	0	0	0	0	1	0				
P3	0	0	1	0	0	0	0	1				
P4	0	1	0	1	1	0	0	0				
P5	0	0	0	1	0	0	0	0				
Sum=	1	2	1	1								

Here,

R1 → 3 instances

R2 → 2 instances

R3 → 1 instance

R4 → 2 instance

P.T.O

(2)

(ii)

Applying Banker's Algorithm:

Process	Allocated Resource				Requested Resource				Available			
	R1	R2	R3	R4	R1	R2	R3	R4	R1	R2	R3	R4
④✓ P1	1	0	0	0	0	1	0	0	2	0	0	0
⑤✓ P2	0	1	0	0	0	0	1	0	2	0	0	1
②✓ P3	0	0	1	0	0	0	0	1	2	0	1	1
③✓ P4	0	1	0	1	1	0	0	0	2	1	1	2
①✓ P5	0	0	0	1	0	0	0	0	3	1	1	2
									3	2	1	2

Here, No deadlock exists.

Safe Sequence: $P_5 \rightarrow P_3 \rightarrow P_4 \rightarrow P_1 \rightarrow P_2$

The system is in a safe state because all processes are complete. Therefore, no deadlock occurs.

Ans: to the question no-02

Given that,

total track 800

number of sectors per track = 150

number of bytes per sector = 512

Time required to move the head from one track to adjacent track is 2ms

Rotation speed 7200 RPM

$$\therefore \text{total number of movements} = \frac{0 + (800 - 1)}{2} = 399.5$$

$$\therefore \text{Seek Time} = (399.5 \times 2) \text{ ms} = 799 \text{ ms}$$

We know that,

7200 RPM means 7200 rotations in 60 sec

$$\therefore 1 \text{ rotation} = \left(\frac{60}{7200}\right) \text{ sec} = 0.00833 \text{ sec} = 8.33 \text{ ms}$$

[$\therefore 1 \text{ sec} = 1000 \text{ ms}$]

$$\therefore \text{Average Rotational Delay (ARD)} = \frac{8.33}{2} \text{ ms} = 4.165 \text{ ms}$$

Reading full track:

150 sectors each has 512 bytes

One rotation data transfer time on reading 150×512 requires 8.33 ms

$$\text{So, for 256 bytes} = \left(\frac{8.33 \times 256}{150 \times 512}\right) \text{ msec} = 0.02777 \text{ msec}$$

(transfer time)

$$\therefore \text{Disk Access Time} = \text{Seek Time} + \text{Average Rotational Delay} + \text{Transfer Time}$$

$$= (799 + 4.165 + 0.02777) \text{ ms}$$

$$= 803.19277 \text{ ms}$$

Ans: Seek time = 799 ms

Average Rotational Delay = 4.165 ms

Transfer Time = 0.02777 ms

Disk Access Time = 803.19277 ms